

Nitrogen and Protein Determination in Rice according to the Kjeldahl Method (Back Titration)

Streamlining essential agro-food testing workflows

The determination of protein in food is a routine procedure for quality assurance and labelling. A simple and fast procedure for protein determination in rice, as described in the AOAC 979.09, is introduced below. The sample is digested with sulfuric acid using the SpeedDigester K-436 or K-439, followed by distillation and back titration with the KjelFlex K-360. The determined protein contents correspond to the labelled values.

Keywords or phrases:

Kjeldahl method, protein determination in rice, nitrogen analysis, back titration, SpeedDigester K-436/K-439, KjelFlex K-360, AOAC 979.09

INTRODUCTION

Protein determination is one of the key analyses performed in the food industry. The samples require digestion with sulfuric acid to convert nitrogen into ammonium sulphate. After conversion to ammonia through the alkalization with sodium hydroxide, the sample is distilled into a sulfuric acid receiver by steam distillation, followed by a back titration with sodium hydroxide solution. The nitrogen content is multiplied by a sample-specific factor (6.25 for rice) to obtain the protein content.

EXPERIMENTAL

Instrumentation: SpeedDigester K-436, K-439, KjelFlex K-360, external dosage device, 848 Titrino plus

Samples: Rice, protein content 7.36 g/100g

Chemicals and Materials: Sulphuric acid conc. 98 %, catalyst, Hg/Se-free, NaOH 32 %, H_2SO_4 0.25 mol/L, standard solution, NaOH 0.5 mol/L, standard solution, neutralization solution for the Scrubber: 600 g sodium carbonate, calcined,

technical, about 2 ml ethanol and a spatula tip of bromthymol blue, diluted to 3 L with distilled water, glycine (assay: 99.7 %)

PROCEDURE

The determination of nitrogen and protein in rice includes the following steps:

- ▶ Homogenization of the sample
- ▶ Digestion of the sample, using SpeedDigester K-436, K-439
- ▶ Distillation of the sample, using KjelFlex K-360
- ▶ Back titration with Metrohm 848 Titrino Plus

Digestion method - glycine (verification of the method)

- ▶ Place approx. 0.25 g glycine in a 300 ml sample tube
- ▶ Add 2 Kjeldahl tablets and a portion of 20 ml of sulfuric acid (98 %)
- ▶ Prepare additional blanks, chemicals without sample
- ▶ Carefully suspend the sample by gently swirling the tube
- ▶ Connect the Scrubber B-414 to the SpeedDigester K-436 or K-439 for absorbing the acid fumes created during digestion
- ▶ Insert the rack containing the samples into the preheated unit

Digest the samples according to the “cereal products” method (K-439) or the parameters listed in Table 1.

Digestion method samples

- ▶ Place approx. 2.5 g of the sample (depends on concentration of protein and organic matrix) in a 300 ml sample tube.
- ▶ Add 2 Kjeldahl tablets and a portion of 25 ml of sulfuric acid (98 %)
- ▶ Prepare additional blanks, chemicals without sample
- ▶ Carefully suspend the sample by gently swirling the tube

- ▶ Connect the Scrubber B-414 to the SpeedDigester K-436 or K-439 for absorbing the acid fumes created during digestion
- ▶ Insert the rack containing the samples into the preheated unit
- ▶ Digest the samples according to the “cereal products” method (K-439) or the parameters listed in Table 1.

	K-439		K-436	
Step	Temp. [°C]	Time [min]	Level	Time [min]
Preheat	480	-	8.5	10
1	480	10	8.5	10
2	550	10	9.5	15
3	490	65	8.5	75
Cooling	-	30	-	30

Table 1. Temperature profile for digestion with the K-436, K-439 K-439 K-436.

NOTE: When the samples are placed in the cooling position it takes approx. 30 min to cool them down; when they are left in the heating chamber it takes at least 60 min.

Distillation and Titration

Distill the samples according to the parameters listed in Table 2. The determination was carried out with a KjellFlex K-360 connected to an external dosage device and a Metrohm 848 Titrino plus.

KjellFlex K-360		Metrohm 848 Titrino plus	
Water	50 ml	Receiving solution ¹	20.0 ml H ₂ SO ₄ 0.25 mol/L
Sodium hydroxide	90 ml	Titration Solution	NaOH 0.5 mol/L
Reaction Time	5 s	Endpoint	pH 4.65
Steam Power	100%	Titration Rate	Optimal
Dist. Time	240 s	Stop Crit.	Drift
Titration Start	240 s	Stop Drift	20 µl/min
Titration Type	Back Titration	Stop Volume	40 ml
Stirrer Sp. Dist.	5	Stop Time	Off
Stirrer Sp. Titr.	7	Filling Rate	Max. mL/min

Table 2. Parameters for distillation with the KjellFlex K-360 and titration.

1) The external dosage device is connected to the port for the manual key button. The volume of the receiving solution must be entered at the external dosage device. For this purpose, refer to the Operation Manual of the dosage device.

Configuration K-360: Signal “ready”: No

Signal “active”: No

Signal “end”: 40 s

CALCULATION

The results are calculated as percentage of nitrogen. In order to calculate the protein content of the sample, the nitrogen content is multiplied with a sample-specific protein factor. The following equations (1), (2), and (3) are used to calculate the results.

$$W_N = \frac{(V_{\text{Blank}} - V_{\text{Sample}}) \cdot C \cdot f \cdot MN}{m_{\text{Sample}} \cdot 1000} \quad (1)$$

$$\%N = W_N \cdot 100 \% \quad (2)$$

$$\%P = W_N \cdot PF \cdot 100 \% \quad (3)$$

W_N : weight fraction of nitrogen

V_{Blank} : mean amount of titrant for the blank [mL]

V_{Sample} : amount of sodium hydroxide titrant for the sample [mL]

c : titrant concentration [mol/L]

f : factor of the titrant (for commercial solutions normally 1.000)

M_N : molecular weight of nitrogen (14.007 g/mol)

m_{Sample} : sample weight [g] (recovery: consider the assay of glycine)

1000 : conversion factor [mL/L]

$\%N$: percentage of weight of nitrogen

$\%P$: percentage of weight of protein

PF : sample-specific protein factor (6.25 for rice)

RESULTS

Digestion with SpeedDigester K-439

Recovery of glycine

The results of the nitrogen determination and recovery in glycine are presented in Table 3. The nominal value of glycine (assay: 99.7 %) is 18.60 % nitrogen. The recoveries are within the specification of 98 – 102 %².

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Glycine	m _{Sample} [g]	V _{Sample} [mL]	%N	Recovery [%]
Sample 1	0.2607	13.028	18.62	100.1
Sample 2	0.2348	13.754	18.51	99.5
Sample 3	0.2339	13.764	18.55	99.7
Sample 4	0.2674	12.908	18.47	99.3
Average	-	-	18.54	99.6
Rsd [%]	-	-	0.35	0.35

Table 3. Results for the recovery of nitrogen in glycine with K-439. The mean blank volume for this sample was 19.959 mL (n = 4).

Protein determination in rice

The results of the determination of nitrogen in rice are presented in Table 4.

Rice	m _{Sample} [g]	V _{Sample} [mL]	%N	%P
Sample 1	2.5409	15.62	1.198	7.49
Sample 2	2.595	15.548	1.192	7.45
Sample 3	2.5755	15.59	1.19	7.44
Sample 4	2.5445	15.658	1.186	7.41
Average	-	-	1.192	7.45
Rsd [%]	-	-	0.43	0.43

Table 4. Results for the determination of nitrogen in rice with K-439 (literature protein content 7.36 %¹).

The mean blank volume for this sample was 19.967 ml (n = 4).

Digestion with SpeedDigester K-436

Recovery of glycine

The results of the nitrogen determination and recovery in

glycine are presented in Table 5. The nominal value of glycine (assay: 99.7 %) is 18.60 % nitrogen. The recoveries are within the specification of 98 – 102 %².

Glycine	m _{Sample} [g]	V _{Sample} [mL]	%N	Recovery [%]
Sample 1	0.2529	13.266	18.54	99.6
Sample 2	0.2607	13.07	18.51	99.5
Sample 3	0.2559	13.164	18.6	100
Sample 4	0.266	12.922	18.53	99.6
Average	-	-	18.54	99.7
Rsd [%]	-	-	0.21	0.21

Table 5. Results for the recovery of nitrogen in glycine with K-436.

The mean blank volume for this sample was 19.959 ml (n = 4).

Protein determination in rice

The results of the determination of nitrogen in rice are presented in Table 6.

Rice	m _{Sample} [g]	V _{Sample} [mL]	%N	%P
Sample 1	2.5727	15.644	1.183	7.39
Sample 2	2.4674	15.832	1.18	7.37
Sample 3	2.702	15.424	1.183	7.39
Sample 4	2.5488	15.702	1.178	7.36
Average	-	-	1.181	7.38
Rsd [%]	-	-	0.21	0.21

Table 6. Results for the determination of nitrogen in rice with K-436 (literature protein content 7.36 %¹).

The mean blank volume for this sample was 19.989 ml (n = 4).

COMPARISON TO STANDARD METHODS

This application note is based on the standard method AOAC 979.09.

	This application note	Standard method	Notes/Impact
Sample weight	2.5 g	0.7 – 2.2 g	In back titration higher contents of nitrogen are needed to obtain a reasonable difference in titrant consumption between blanks and samples.
Catalyst	10 g Tablets cont. - 47.7 % K ₂ SO ₄ - 47.7 % Na ₂ SO ₄ - 2.8 % TiO ₂ - 1.8 % CuSO ₄	15 g K ₂ SO ₄ + 0.7 g HgO	Easy to handle especially in routine analytics. The choice of catalyst does not influence the result. No toxic Hg.
Water	50 mL	200 mL	The K-360 generates steam in a separate vessel; therefore it is not necessary to add such a high amount of water to the digested sample as described in the standard method.

Table 7: Differentiation to the standard method

CONCLUSION

The determination of nitrogen and protein in rice using the SpeedDigester K-436, K-439, and KjelFlex K-360 provides reliable and reproducible results that correspond to the literature values with low relative standard deviations. There are no differences between the results obtained with the K-436 and the K-439 respectively.

The digestion time is very fast: 85 min for the K-439 and 100 min for the K-436. The recoveries with glycine were 99.6 % (K-439) and 99.7 % (K-436) respectively and are within the specification of 98 – 102 %.

REFERENCES

1. Souci Fachmann Kraut, CRC Press, 7th edition, 2008
2. Application Note 001-437_370-03C: Operational Quality Check Procedure

AOAC 979.09

Operation manual of SpeedDigester K-425 / K-436

Operation manual of SpeedDigester K-439

Operation manual of Scrubber B-414

Operation manual of KjelFlex K-360

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KjelMaster K-375 & KjelSampler K-376 / K-377

